DEPARTMENT OF COMPUTER ENGINEERING
MIT Academy of Engineering, Alandi (MITAOE)

AI-POWERED SMART HEALTHCARE ASSISTANT SYSTEM

(Medicare)

TEST-DRIVEN DEVELOPMENT PLAN

Practical 5 — Software Engineering

Version 1.0 | Academic Year 2025–2026

Project Title	AI-Powered Smart Healthcare Assistant System (Medicare)
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Version	1.0 – Final Submission
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Development Model	Agile – 3 Sprints + Integration & Deployment Phases
Team Members	Anushka Jadhav, Aditi Namde, Vedant Ingle, Shravani Karhale, Harshad Gore, Devesh Kumar
Client	Healthcare Clinics / Hospitals
Department	Computer Engineering, MITAOE

1. Purpose

This Test-Driven Development (TDD) Plan establishes a structured, risk-anchored approach to designing, writing, and executing tests for the AI-Powered Smart Healthcare Assistant System (Medicare). Unlike a conventional test plan appended after development, every test in Medicare is written before the corresponding production code, following the Red–Green–Refactor cycle mandated by TDD.

This document is directly derived from the Medicare Software Requirements Specification (SRS v1.0), the Healthcare Risk Management Matrix, and the project estimation documents (WBS, Cost Estimation). Each test case is anchored to a specific Risk ID from the risk register. No feature is implemented without a corresponding failing test.

The plan covers five test phases: Unit Testing, Integration Testing, Security Testing, Performance Testing, and User Acceptance Testing (UAT). All phases are mapped to the three Agile sprints and post-sprint integration/deployment phases, running from 05 January 2026 to 25 April 2026.

By implementing this plan, the Medicare project team commits to:

- Writing failing tests (Red) before any production code for every functional requirement.
- Writing the minimum code to pass each test (Green), then refactoring.
- Anchoring every test to a Risk ID — if no risk entry exists, no code is written.
- Running the full test suite on every push to the main branch via GitHub Actions CI/CD.
- Achieving 100% pass rate on all 9 UAT acceptance criteria (TC-01 to TC-09) at the final UAT milestone.

2. Scope

2.1 In-Scope Test Categories

- Authentication & Access Control Tests: JWT handling, RBAC enforcement, OTP verification, session management, privilege escalation prevention.
- Business Logic Tests: Appointment slot locking, double-booking prevention, prescription expiry, OCR confidence thresholds, chatbot safety guardrails.
- Security Tests: RLS/RBAC policy enforcement, encryption at rest (AES-256), TLS enforcement, XSS/injection protection, brute-force lockout, malicious file upload prevention.
- Performance Tests: API response times (p95 ≤ 2s), 500 concurrent users, prescription upload within 5 seconds, doctor search under load.
- Integration Tests: API-to-database flows, Twilio/Vapi/Gemini API integration, notification delivery, audit logging, n8n automation workflows.
- User Acceptance Tests: TC-01 through TC-09 — covering all core patient, doctor, and admin workflows with real users.

2.2 Out-of-Scope

- Phase 2 features explicitly deferred post-submission: video consultation, online payment, mobile native apps (iOS/Android), WhatsApp integration, blockchain medical records.
- Third-party infrastructure reliability (Twilio/Vapi/Gemini API uptime) — mitigated by fallback architecture and circuit breakers, not by automated tests.
- Physical clinic or hospital counter hardware — addressed by deployment documentation, not automated tests.

3. TDD Core Cycle

All development in Medicare follows the Red–Green–Refactor cycle. The cycle is triggered by a Risk ID, not a feature request. The following table defines each stage in the Medicare context.

Stage	Colour Signal	Medicare Definition
Stage 1 — Red	**FAIL**	Write a failing test derived from the SRS functional requirement and its associated Risk ID. For example, test the slot-locking mechanism for R-04 edge case 'two patients book the same slot simultaneously — only one must succeed'. No production API or SQL code exists yet.
Stage 2 — Green	**PASS**	Write the minimum production code to make the test pass. This means writing only the atomic slot-locking transaction, the RBAC middleware, the file MIME-type validator, or the chatbot disclaimer injector — exactly enough to satisfy the test assertion.
Stage 3 — Refactor	**STILL PASS**	Apply production-quality hardening: ACID transactions and unique constraints (R-04), AES-256 encryption and RLS enforcement (R-01, R-02, R-12), TLS 1.3 enforcement (R-02), rate-limiting middleware (R-03), OCR confidence gating (R-08), and chatbot safety rules (R-15). Re-run the full suite after every refactor to confirm no regression.

Key principle: Every TDD iteration is anchored to a specific Risk ID from the Medicare Risk Management Matrix. If a feature request has no corresponding risk entry, a new risk item must be added to the register before the test is written.

4. Risk-to-Test Mapping

The following matrix maps each risk from the Medicare Risk Management Matrix to its corresponding test phase, test IDs, and post-mitigation risk level. This table confirms that every Extreme and High risk has at least one automated test guarding against it.

15	6	5	8	5
Total Risks	EXTREME Risks Mitigated by Tests	HIGH Risks Monitored in CI/CD	Core Modules Under Test	Test Phases Across Sprints

Risk ID	Risk Name	Department	Pre-Mitig.	Post-Mitig.	Test Priority
R-01	Unauthorized access to patient data (weak RBAC / privilege escalation)	Security / Backend	EXTREME	MEDIUM	CRITICAL
R-02	Data breach due to insecure storage of prescriptions / medical history	Security / Storage	EXTREME	MEDIUM	CRITICAL
R-03	Broken authentication (token hijacking / weak JWT handling)	Backend / Auth	HIGH	LOW	HIGH
R-04	Appointment double-booking due to race conditions	Backend / Appointments	HIGH	LOW	HIGH
R-05	Wrong slot availability shown (cached / stale data)	Backend + UI	MEDIUM	LOW	MEDIUM
R-06	Prescription upload failure (network interruption / timeout)	Frontend / Upload	MEDIUM	LOW	MEDIUM
R-07	Malicious file upload (malware in PDF / image)	Security / Upload	EXTREME	MEDIUM	CRITICAL
R-08	OCR extraction errors causing wrong medicine details	AI / OCR Module	HIGH	MEDIUM	HIGH
R-09	Automated follow-up call fails (Twilio / Vapi outage)	Automation / Integrations	MEDIUM	LOW	MEDIUM
R-10	Notification not delivered (SMS / Email gateway issue)	Notification Service	MEDIUM	LOW	MEDIUM
R-11	Doctor verification fraud (fake license / certificates)	Admin / Verification	EXTREME	MEDIUM	CRITICAL
R-12	Unauthorized medical history access (privacy breach)	Security / Data Access	EXTREME	MEDIUM	CRITICAL
R-13	Data loss due to missing backups or accidental deletion	Database / Storage	HIGH	LOW	HIGH
R-14	Performance degradation under peak load (booking / upload spikes)	Backend / Infrastructure	HIGH	MEDIUM	HIGH
R-15	AI chatbot gives unsafe / incorrect medical advice	AI Assistant	EXTREME	MEDIUM	CRITICAL

5. Phase 1 — Unit Testing

Sprint: 1–2

Tools: Jest (Node.js), React Testing Library, MongoDB In-Memory Server

Responsible: Devesh Kumar (Auth/RBAC), Harshad Gore (Appointment logic), Vedant Ingle (OCR/AI), Shravani Karhale (Upload/Storage)

Unit tests cover all pure business logic functions in isolation. Each test is written before the corresponding API handler, SQL trigger, or JavaScript utility is implemented. All tests must pass before code is merged to the main branch.

5.1 RBAC and JWT Authentication — validateRBAC() / verifyJWT()

Risk R-01, R-03 (EXTREME/HIGH pre-mitigation): The RBAC middleware and JWT verification must correctly enforce role-based access and reject tampered or expired tokens before any route handler executes.

TC ID	Input Condition	Expected Outcome	Risk
UT-A-01	Patient JWT → access /api/admin/* endpoint	HTTP 403 Forbidden — role check fails at middleware	R-01 EXTREME
UT-A-02	JWT with role manually changed to 'admin' (invalid signature)	HTTP 401 Unauthorized — signature mismatch detected	R-03 HIGH
UT-A-03	Expired JWT submitted (past exp claim)	HTTP 401 Unauthorized — token expired error returned	R-03 HIGH
UT-A-04	Doctor token used to access unassigned patient's records	HTTP 403 Forbidden — RBAC patient-assignment check fails	R-01 EXTREME
UT-A-05	6th failed login attempt within 15-minute window	HTTP 429 Too Many Requests — account temporarily locked	R-03 HIGH
UT-A-06	Session active for 25 hours with no activity (timeout = 24h)	JWT invalidated — user forced to re-authenticate	R-03 HIGH
UT-A-07	Logout called → reuse previous access token on next request	HTTP 401 Unauthorized — token invalidated on logout	R-03 HIGH

5.2 Appointment Slot Locking — bookSlot() / checkSlotAvailability()

Risk R-04 (HIGH pre-mitigation): The slot reservation function must use atomic DB transactions and a unique constraint on (doctorId, appointmentDate, appointmentTime) to prevent any double-booking regardless of concurrent requests.

TC ID	Input Condition	Expected Outcome	Risk
UT-B-01	Single patient books available slot — no contention	Booking created; slot status updated to RESERVED	R-04 HIGH
UT-B-02	Two concurrent booking requests for the exact same slot	Exactly one booking confirmed (HTTP 200); other returns HTTP 409 Conflict	R-04 HIGH
UT-B-03	Booking started but abandoned — slot held beyond timeout	Slot reverts to AVAILABLE after lock timeout (configurable TTL)	R-04 HIGH
UT-B-04	DB unique constraint violation for duplicate (doctorId, slot)	DB throws unique constraint error; caught and returned as 409	R-04 HIGH
UT-B-05	Idempotent retry with same idempotency key	Same confirmation returned; no duplicate appointment created	R-04 HIGH
UT-B-06	Patient cancels appointment > 2 hours before scheduled time	Appointment cancelled; slot released to AVAILABLE	R-04 HIGH

5.3 File Upload Validation — validateUpload() / sanitizeFile()

Risk R-07 (EXTREME pre-mitigation): The upload service must validate file type, size, and MIME type before any storage operation. Malicious files must be blocked at the entry point, not just after storage.

TC ID	Input Condition	Expected Outcome	Risk
UT-F-01	Upload valid PDF prescription (≤ 10 MB)	File accepted; upload proceeds to encryption and storage	R-07 EXTREME
UT-F-02	Upload .exe file renamed to .pdf (spoofed extension)	HTTP 400 — server reads real MIME type; file rejected	R-07 EXTREME
UT-F-03	Upload file of 11 MB (over 10 MB limit)	HTTP 413 Payload Too Large — upload rejected immediately	R-07 EXTREME
UT-F-04	Upload PDF with embedded JavaScript content	PDF sanitised; embedded scripts stripped before storage	R-07 EXTREME
UT-F-05	Upload EICAR antivirus test file (standard AV test string)	Antivirus scan blocks file; alert logged; HTTP 400 returned	R-07 EXTREME
UT-F-06	Prescription stored — check AES-256 encryption of file bytes	Stored bytes are ciphertext; decrypted only via authenticated API	R-02 EXTREME

5.4 OCR Confidence Gating — extractPrescription() / flagLowConfidence()

Risk R-08 (HIGH pre-mitigation): The OCR module must apply a confidence threshold to extracted fields. Any field below the threshold must be flagged for manual review, and the original prescription image must always be preserved.

TC ID	Input Condition	Expected Outcome	Risk
UT-O-01	Clear prescription image — all OCR fields high confidence	All fields extracted; none flagged; no manual review required	R-08 HIGH
UT-O-02	Blurry prescription — dosage field below confidence threshold	Dosage field flagged in UI with 'Please verify' indicator	R-08 HIGH
UT-O-03	OCR reads '100mg' vs actual '1000mg' — low confidence	Field highlighted for correction; original image retained in DB	R-08 HIGH
UT-O-04	User manually corrects OCR-extracted medication name	Corrected value saved; original OCR output preserved in audit log	R-08 HIGH
UT-O-05	Batch OCR on 20 diverse prescription samples	Accuracy ≥ 85% on medication name and dosage fields across sample set	R-08 HIGH

5.5 AI Chatbot Safety Rules — processQuery() / injectDisclaimer()

Risk R-15 (EXTREME pre-mitigation): The chatbot response pipeline must apply safety guardrails before returning any response to a patient. Specific dosage advice must be refused, and all medical responses must include a disclaimer.

TC ID	Input Condition	Expected Outcome	Risk
UT-C-01	Patient asks: 'What dosage of metformin should I take?'	Chatbot refuses specific dosage; directs to prescribing doctor	R-15 EXTREME
UT-C-02	Patient asks: 'Should I stop taking my blood pressure pills?'	Chatbot advises not to stop medication without consulting doctor	R-15 EXTREME
UT-C-03	Any medical response returned by Gemini API	Disclaimer appended: 'This is general information. Consult your doctor.'	R-15 EXTREME
UT-C-04	Gemini API returns response with confidence below threshold	Fallback safe message returned; specific advice not shown	R-15 EXTREME
UT-C-05	Patient types: 'I need to speak to a human'	Chatbot provides escalation path / human support contact details	R-15 EXTREME
UT-C-06	All chatbot interactions logged to ChatLogs collection	Query, response, intent, confidence, and timestamp stored in DB	R-15 EXTREME

6. Phase 2 — Integration Testing

Sprint: 3

Tools: Jest, Supertest (Express.js API testing), Postman / Newman, MongoDB In-Memory Server
Responsible: Aditi Namde (API flows), Anushka Jadhav (Notifications), Devesh Kumar (Auth + DB)

Integration tests validate complete request-to-response flows through multiple system layers: from HTTP request through the Express.js API Gateway, through business logic services, into MongoDB, and back. These tests are derived from the sequence diagrams and API endpoint definitions in the SRS Appendix C, covering the full Book Appointment, Upload Prescription, Doctor Verification, and Automation flows.

6.1 Appointment Booking Full Flow

TC ID	Phase	Scenario	Expected Outcome	Risk
IT-B01	Phase 1	GET /api/doctors with specialty filter on dashboard init	Returns filtered doctor list with {id, name, specialty, rating, available} in < 2s	R-05, R-14
IT-B02	Phase 2	POST /api/appointments with valid doctorId and slot	200 OK; appointment created; confirmation SMS and email triggered	R-04, R-09
IT-B03	Phase 2	POST /api/appointments — slot just taken by another user	409 Conflict returned; alternate slots suggested in response	R-04 HIGH
IT-B04	Phase 3	50 simultaneous POST /appointments for the same slot	Exactly 1 appointment created; 49 return 409 (ACID verified)	R-04 HIGH
IT-B05	Phase 3	DELETE /api/appointments/:id — patient cancels > 2 hrs before	Appointment cancelled; doctor notified via dashboard + SMS	R-04, R-10
IT-B06	Phase 3	POST /api/appointments/:id/reschedule with new slot	Appointment rescheduled; old slot released; confirmations sent	R-04, R-10

6.2 Prescription Upload and Medical History Flow

TC ID	Scenario	Expected Outcome	Risk
IT-P01	POST /api/prescriptions/upload — valid PDF ≤ 10 MB by authenticated patient	File stored encrypted; OCR triggered; metadata saved in Prescriptions collection	R-02, R-07
IT-P02	POST /api/prescriptions/upload — network drops mid-upload at 50%	Upload resumes from byte offset on retry; no duplicate file stored	R-06 MEDIUM
IT-P03	GET /api/patients/medical-history — authenticated patient	Full chronological record returned; allergies and chronic conditions highlighted	R-12 EXTREME

TC ID	Scenario	Expected Outcome	Risk
IT-P04	GET /api/medical-history/:patientId — doctor not assigned to patient	HTTP 403 Forbidden; access denied at RBAC layer	R-12 EXTREME
IT-P05	GET /api/medical-history/:patientId — assigned doctor during valid access grant	Records returned; audit log entry created with doctorId + timestamp	R-12 EXTREME
IT-P06	POST /api/prescriptions/:id/share — patient shares with new doctor	Access granted for specified doctor; notification sent to doctor	R-12, R-10

6.3 Doctor Verification Workflow

TC ID	Scenario	Expected Outcome	Risk
IT-V01	POST /api/doctors/verify/submit — doctor uploads license and certificates	Verification request created in DB; admin notified via dashboard	R-11 EXTREME
IT-V02	Patient attempts to book appointment with unverified doctor	Booking rejected; unverified doctor not shown in search results	R-11 EXTREME
IT-V03	POST /api/admin/doctors/:id/verify — admin approves doctor	Status updated to VERIFIED; verified badge displayed; doctor notified	R-11 EXTREME
IT-V04	POST /api/admin/doctors/:id/reject — admin rejects with reason	Status updated to REJECTED; reason stored in DB; doctor notified via email	R-11 EXTREME
IT-V05	Doctor verificationDate = 11.5 months ago — annual check triggers	Re-verification reminder sent to doctor automatically via n8n workflow	R-11 EXTREME

6.4 Automated Follow-Up and Notification Flow

TC ID	Scenario	Expected Outcome	Risk
IT-N01	n8n workflow triggers medication reminder at scheduled time	Twilio call initiated; call status logged in Reminders collection	R-09 MEDIUM
IT-N02	Twilio returns HTTP 503 (service outage) during follow-up call	Circuit breaker opens; fallback SMS sent; patient notified via alternate channel	R-09 MEDIUM

TC ID	Scenario	Expected Outcome	Risk
IT-N03	Failed call — retry with exponential backoff up to 3 attempts	3 retry attempts logged at t+5min, t+15min, t+30min intervals	R-09 MEDIUM
IT-N04	Patient has opted out of automated calls in preferences	No call initiated; opt-out status checked before any automation trigger	R-09 MEDIUM
IT-N05	Appointment confirmation SMS not delivered (gateway bounce)	Delivery failure logged; retry via email fallback; status updated in DB	R-10 MEDIUM

7. Phase 3 — Security Testing

Sprint: Integration & Security Phase (after Sprint 3)

Tools: OWASP ZAP, Jest (JWT + RBAC tests), custom security test scripts, Postman security collection

Responsible: Vedant Ingle (encryption + storage), Omkar-style: Harshad Gore (RBAC + JWT), Anushka Jadhav (audit + compliance)

Security tests are executed before deployment to the production cloud environment (AWS / Docker / Kubernetes). All RBAC and JWT tests run automatically on every push to the main branch via GitHub Actions. Any failure blocks the merge. Encryption (R-02), access breach prevention (R-01, R-12), malicious upload blocking (R-07), and data deletion protection (R-13) are tested at both application and database levels.

TC ID	Risk	Scenario	Expected Outcome
SEC-01	R-01	Patient role submits request to GET /api/admin/users	HTTP 403 Forbidden — RBAC middleware blocks at route level
SEC-02	R-01	Doctor token attempts access to non-assigned patient record	HTTP 403 Forbidden — patient-assignment check blocks access
SEC-03	R-02	Direct access to MongoDB — inspect medical history documents	Diagnosis, medications, and vitals fields are encrypted ciphertext
SEC-04	R-02	HTTP (non-TLS) request submitted to any API endpoint	Connection refused or force-redirected to HTTPS
SEC-05	R-03	Attacker replays stolen JWT from previous session	HTTP 401 — token blacklisted on logout; replay rejected
SEC-06	R-03	CI/CD push — RBAC middleware test fails	Merge blocked automatically; status shown in GitHub Actions
SEC-07	R-07	XSS payload in prescription metadata field (e.g., fileName)	Input sanitised; payload not executed; stored as escaped string
SEC-08	R-07	SQL/NoSQL injection in doctor search query parameter	Query sanitised; injection string treated as literal search term

TC ID	Risk	Scenario	Expected Outcome
SEC-09	R-11	Doctor status changed to VERIFIED via patient or doctor API endpoint	HTTP 403 Forbidden — only /api/admin/doctors/:id/verify can change status
SEC-10	R-12	Audit log reviewed after doctor accesses patient medical history	Log entry: doctorId, patientId, records accessed, timestamp recorded
SEC-11	R-12	Doctor accesses 50+ patient records within 1 minute	Security alert triggered; admin notified; access rate-limited
SEC-12	R-13	Patient attempts DELETE on /api/medical-history/:id	HTTP 403 Forbidden — hard-delete not permitted for patient role
SEC-13	R-13	Admin attempts hard DELETE — no soft-delete override	Confirmation required; soft-delete flag set (is_deleted = true); record retained

8. Phase 4 — Performance Testing

Sprint: Integration & Performance Phase (after Sprint 3)

Tools: k6, Artillery, Apache JMeter, Browser Performance API

Responsible: Harshad Gore (load testing), Devesh Kumar (DB concurrency), Shravani Karhale (upload performance)

Performance tests validate the core SLAs defined in the Medicare SRS Section 5.1: API response time p95 ≤ 2 seconds, page load ≤ 3 seconds, 500 concurrent users supported, and prescription upload ≤ 5 seconds for 10 MB files. Tests run against a staging environment populated with realistic data (500 patients, 50 doctors, 1,000 appointments).

8.1 Appointment Booking Performance

TC ID	Scenario	Pass Criterion	Risk
PT-A-01	100 sequential doctor search requests (specialty filter, n = 50 doctors)	All results returned in p95 ≤ 2,000ms	R-14 HIGH
PT-A-02	Single appointment booking (patient → slot confirm → DB write)	End-to-end response in < 2,000ms	R-14 HIGH
PT-A-03	50 simultaneous POST /appointments — concurrency stress	Zero duplicate bookings; all ACID-compliant; correct balances	R-04, R-14
PT-A-04	500 concurrent virtual users for 10-minute sustained load	p95 ≤ 2,000ms; error rate < 1%; no service crash	R-14 HIGH
PT-A-05	Rate limiting check — 200 API requests per minute from single IP	HTTP 429 returned after 100th request; limit enforced	R-14 HIGH

8.2 Prescription Upload Performance

TC ID	Scenario	Pass Criterion	Risk
PT-P-01	Upload 10 MB prescription PDF under normal load (1 concurrent user)	Upload + encryption + storage completes in ≤ 5 seconds	R-14 HIGH
PT-P-02	Upload 10 MB prescription PDF under 50 concurrent upload requests	All uploads complete correctly; no file corruption; p95 ≤ 8 seconds	R-14 HIGH
PT-P-03	Resumable upload — 10 MB file, network interrupted at 50%	Upload resumes from byte offset; total time within 10 seconds	R-06 MEDIUM
PT-P-04	OCR processing triggered post-upload for 10 different prescriptions	OCR completes per prescription in ≤ 10 seconds; results stored in DB	R-08, R-14

8.3 System-Level Performance

TC ID	Scenario	Pass Criterion	Risk
PT-S-01	Page load for patient dashboard (cold start)	Dashboard renders in ≤ 3 seconds on standard broadband connection	R-14 HIGH
PT-S-02	MongoDB query — fetch patient medical history (20 records)	Query completes in ≤ 1,000ms with correct index usage	R-14 HIGH
PT-S-03	Automated daily backup process	Backup completes in ≤ 30 minutes; backup file integrity verified	R-13 HIGH
PT-S-04	Disaster recovery simulation — full restore from backup	System operational within 4-hour Recovery Time Objective (RTO)	R-13 HIGH
PT-S-05	AI chatbot query response (Gemini API call + safety processing)	Response returned to patient in ≤ 3 seconds for standard query	R-15 EXTREME

9. Phase 5 — User Acceptance Testing (UAT)

Sprint: Sprint 5 — Final Week (April 2026)
Conducted with: Healthcare Clinic Representative (Client), Full project team present
Responsible: Anushka Jadhav (lead), all 6 members

UAT is conducted with the client representative at a designated healthcare clinic. The 9 test scenarios below are executed with real users (a patient, a doctor, and an admin) against the production-equivalent deployment instance. Test devices include Android and iOS smartphones and a laptop browser.

UAT sign-off criteria: All 9 scenarios below must pass. Any failure triggers escalation to the project supervisor before deployment is approved.

TC ID	Scenario	Actor	Expected Outcome	Risk
TC-01	Patient attempts to book an appointment with an unverified doctor	Patient	Booking rejected; unverified doctor not visible in search results	R-11
TC-02	Patient books appointment with verified doctor — slot selected and confirmed	Patient	Appointment created; confirmation SMS and email received within 30 seconds	R-04, R-10
TC-03	Two patients attempt to book the same doctor slot at the same time	2 Patients	Exactly one booking succeeds; second patient receives conflict message and alternate slots	R-04
TC-04	Patient uploads prescription via drag-and-drop interface (PDF, 5 MB)	Patient	Upload succeeds; progress bar shows 0–100%; prescription visible in dashboard	R-06, R-07
TC-05	Doctor accesses medical history of assigned patient	Doctor	Records displayed in chronological order; allergies highlighted; access logged in audit trail	R-12
TC-06	Doctor attempts to access medical history of non-assigned patient	Doctor	Access denied — HTTP 403 shown in dashboard; no records exposed	R-12
TC-07	Admin approves doctor verification — documents reviewed and approved	Admin	Doctor status updated to VERIFIED; verified badge visible; doctor receives approval notification	R-11
TC-08	Patient asks chatbot: 'What is the dosage of paracetamol for fever?'	Patient	Chatbot provides general information with disclaimer; does not prescribe specific dosage	R-15
TC-09	Automated follow-up call scheduled — Twilio initiates call to patient	System + Patient	Patient receives call at scheduled time; call outcome logged; SMS fallback works if call fails	R-09

10. CI/CD Integration and Test Gate

All automated tests run inside the GitHub Actions CI/CD pipeline on every push to the main branch. The pipeline enforces the following gate: a pull request that fails any test suite cannot be merged. This ensures the main branch is always in a deployable, test-passing state.

Pipeline Stage	Tests Included	Sprint Active	Trigger
Lint + Unit	UT-A (Auth/RBAC), UT-B (Slot locking), UT-F (File validation),	Sprints 1–3	Every push to main

Pipeline Stage	Tests Included	Sprint Active	Trigger
	UT-O (OCR), UT-C (Chatbot safety)		
Integration	IT-B (Booking flows), IT-P (Prescription + history), IT-V (Doctor verification), IT-N (Notifications)	Sprint 3 onwards	Every push to main
Security (RBAC + Encryption)	SEC-01 to SEC-13 (RBAC, JWT, encryption, audit)	Sprint 3 onwards	Every push to main — merge blocked on failure
Performance	PT-A-01 to PT-A-05, PT-P-01 to PT-P-04, PT-S-01 to PT-S-05	Integration Phase only	Scheduled nightly on staging
UAT Checklist	TC-01 to TC-09 (manual run with client representative)	Sprint 5 only	Final milestone — client sign-off required

11. TDD Sprint Schedule

Sprint	Weeks	TDD Activity	Responsible	Test Suites Written
Sprint 1 Core Backend & Auth	W1–W4 (Jan 5–Feb 1)	Write failing tests for RBAC enforcement, JWT handling, OTP verification, and bcrypt password checks before any auth code	Devesh Kumar Harshad Gore	UT-A-01 to UT-A-07 (Auth / RBAC unit tests)
Sprint 2 Appointments & Prescriptions	W5–W9 (Feb 2–Mar 1)	Write failing tests for slot locking, double-booking prevention, file validation, encryption, and OCR confidence gating before dashboard or upload code	Harshad Gore Shravani Karhale Vedant Ingle	UT-B-01 to UT-B-06 UT-F-01 to UT-F-06 UT-O-01 to UT-O-05
Sprint 3 AI, Automation & Admin	W9–W11 (Mar 2–Mar 22)	Write failing tests for chatbot safety rules, n8n automation triggers, notification flows, and doctor verification workflow before any AI or admin code	Vedant Ingle Anushka Jadhav Aditi Namde	UT-C-01 to UT-C-06 IT-B-01 to IT-B-06 IT-P-01 to IT-P-06 IT-V-01 to IT-V-05 IT-N-01 to IT-N-05
Integration & Security Phase	W11–W13 (Mar 22– Apr 5)	Security test suite (RBAC, JWT replay, encryption, audit trail) and performance load tests against staging environment	Harshad Gore Anushka Jadhav Vedant Ingle	SEC-01 to SEC-13 PT-A-01 to PT-A-05 PT-P-01 to PT-P-04 PT-S-01 to PT-S-05

Sprint	Weeks	TDD Activity	Responsible	Test Suites Written
Sprint 5 Deployment & UAT	W14–W16 (Apr 6–Apr 25)	UAT execution with client representative, final risk register review, CI/CD gate verification, and submission documentation	Full Team + Client Rep.	TC-01 to TC-09 (UAT sign-off)

12. Conclusion

This Test-Driven Development Plan ensures that every significant risk identified in the Medicare Risk Management Matrix is addressed by at least one automated, version-controlled test before the corresponding production code exists. By anchoring the TDD cycle directly to Risk IDs, the team guarantees that no risk is mitigated on paper alone — it must be mitigated in code.

The five-phase test architecture — Unit, Integration, Security, Performance, and UAT — provides full coverage from individual API handler functions and business logic utilities up to client-facing acceptance criteria. The GitHub Actions CI/CD gate ensures that the test suite is continuously enforced across all sprints, making regression a detectable and blockable event rather than a silent failure.

The plan will be reviewed at each sprint milestone. If new risks are identified during development, corresponding test cases are added before any new feature code is written. The final test register, alongside all passing CI/CD run logs, will be included in the complete project documentation bundle submitted on 25 April 2026.

Document Prepared By	Reviewed By	Approved By
Team Medicare (Anushka Jadhav, Aditi Namde, Vedant Ingle, Shravani Karhale, Harshad Gore, Devesh Kumar)	Project Supervisor Department of Computer Engineering MITAOE, Alandi	Client Representative Date: _____

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